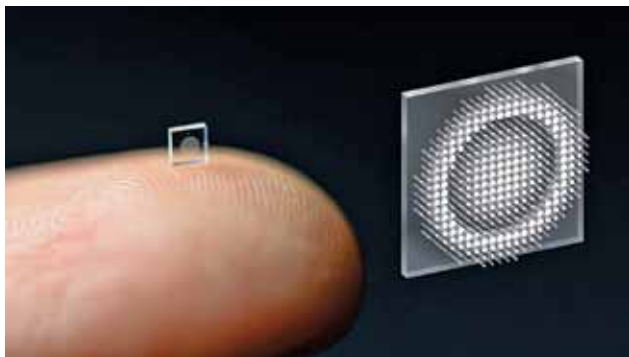


Salt-grain size camera captures high-quality images

Researchers at Princeton University and the University of Washington have devised an ultracompact camera, the size of a coarse grain of salt, that can produce crisp, full-colour images on par with a conventional compound



This ultracompact camera the size of a coarse grain of salt relies on a technology called a metasurface, which is studded with 1.6 million cylindrical posts and can be produced much like a computer chip (Credit: <https://engineering.princeton.edu>)

camera lenses 500,000 times larger in volume. Enabled by a joint design of the camera's hardware and computational processing, the system could enable minimally invasive endoscopy with medical robots to diagnose and treat diseases and improve imaging for other robots with size and weight constraints. Arrays of thousands of such cameras could be used for full-scene sensing, turning surfaces into cameras.

AI-designed self-replicating living robots created

Scientists at the University of Vermont (UVM), in collaboration with Tufts University and Harvard University, have

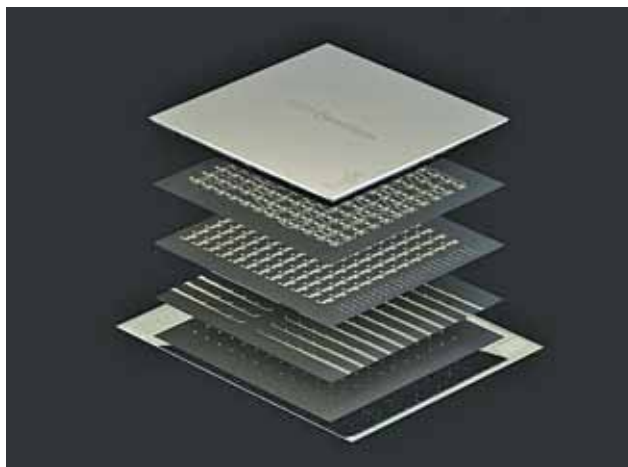


Made from frog cells, these computer-designed organisms gather single cells inside a Pac-Man-shaped 'mouth'—and release Xenobot 'babies' that look and move like themselves. Then the offspring go and do the same—over and over (Credit: www.uvm.edu/news)

discovered an entirely new form of biological reproduction and applied their discovery to create the first-ever, self-replicating living robots called Xenobots. These computer-designed and hand-assembled organisms, inspired by *Xenopus* frog cells, can swim out into their tiny dish, find single cells, gather hundreds of them together, and assemble 'baby' Xenobots inside their Pac-Man-shaped mouth that, a few days later, become new Xenobots that look and move just like themselves. Scientists believe the development will help them create new technological solutions like deploying living machines to pull microplastics out of waterways or build new medicines.

First quantum processor that no computer can match

IBM's new 127-quantum bit (qubit) Eagle processor is a breakthrough in tapping into the massive computing potential of devices based on quantum physics. It can contain



127 qubits on a single IBM quantum processor for the first time with breakthrough packaging technology (Credit: <https://newsroom.ibm.com>)

more than 100 operational and connected qubits. The increased qubit count will allow users to explore problems at a new level of complexity when undertaking experiments and running applications, such as optimising machine learning or modeling new molecules and materials for use in areas spanning from the energy industry to the drug discovery process. Eagle is the first IBM quantum processor whose scale makes it impossible for a classical computer to reliably simulate.

The first battery that is both flexible and washable

University of British Columbia (UBC) researchers have created what could be the first battery that is both flexible and